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# Evgenii Nikishin

## Education

2021–Now Ph.D. student in Computer Science.
**University of Montreal, Mila**
Advisors: Pierre-Luc Bacon, Aaron Courville

2019–2020 Ph.D. student in Operations Research.
**Cornell University**, transferred to Mila

2017–2019 M.Sc. in Computer Science, with honors (GPA: 9.2/10.0).
**Higher School of Economics** (HSE) and
**Skolkovo Institute of Science and Technology**
Advisor: Dmitry Vetrov

2013–2017 B.Sc. in Computer Science, with honors (GPA: 4.9/5.0).
**Lomonosov Moscow State University**
Advisor: Alexander D'yakonov

## Experience

May 2020–Aug 2020 Research Intern (remote), **University of Montreal, Mila**.
Control-oriented model-based RL. Hosts: Pierre-Luc Bacon, Yoshua Bengio

Sep 2017–Aug 2019 Research Assistant, **Bayesian Methods Research Group**.
A part of **Samsung AI Lab**
Impoving stability and transfer in deep RL. Supervisor: Dmitry Vetrov

Jul 2018–Aug 2018 Research Intern, **ETH Zürich**.
Summer Research Fellowship (a total of 16 fellows among 1k applicants). Host: Gunnar Rätsch

Jul 2016–Dec 2016 Research & Development Intern, **Kaspersky Lab**.
Improved feature extraction from textual representations of files for malicious software detection

## Selected awards

2020 Mitacs-Mila short-term research grant.

2019 Cornell graduate fellowship.

2019 NeurIPS 2019 travel award.

2019 MLSS 2019 full travel grant.

2017 Fourth place in DeepHack RL Hackathon.

2016 Russian state academic scholarship.

2016 Second place in Yandex Chatbot Hackathon.

2016 Winner of the blitz tournament at Yandex Data Analysis Night.

2015 Bronze medal in Driver Telematics Analysis competition on Kaggle.

## Publications

2021 Control-Oriented Model-Based Reinforcement Learning with Implicit Differentiation.
**Evgenii Nikishin**, Romina Abachi, Rishabh Agarwal, Pierre-Luc Bacon
A model learning method for RL that directly optimizes the sum of rewards instead of likelihood, a proxy to the agent's objective
ArXiv Preprint. Code

2019 Unsupervised Domain Adaptation with Shared Latent Dynamics for Reinforcement Learning.
**Evgenii Nikishin**, Arsenii Ashukha, Dmitry Vetrov
Domain adaptation via learning shared dynamics in a latent space and adversarial matching of the latent representations of states
Bayesian Deep Learning workshop, NeurIPS 2019. Code

2018 Improving Stability in Deep Reinforcement Learning with Weight Averaging.
**Evgenii Nikishin**, Pavel Izmailov, Ben Athiwaratkun, Dmitrii Podoprikhin,
Timur Garipov, Pavel Shvechikov, Dmitry Vetrov, Andrew Gordon Wilson
Averaging weights during training to reduce the variance of policy gradient estimates
Uncertainty in Deep Learning workshop, UAI 2018

## Other research projects

2018 **Reinforcement Learning for Personal Treatment**.
Off-policy policy gradient methods for treatment of patients in intensive care units

2017 **Hierarchical Imitation Learning**.
Generative Adversarial Imitation Learning for option-augmented policies

2017 **Stable Matrix Approximation via Autoencoders**.
Autoencoder-based architecture for recommender systems with stability guarantees. Code

2016 **Community Detection Methods in Social Graphs**.
Review and comparison of overlapping and non-overlapping community detection algorithms
Accepted to Lomonosov 2017 conference. Code

## Service

Reviewer: NeurIPS 2021, ICML 2021, ICLR 2021 (outstanding reviewer), AAAI 2021, ICML 2020 BIG workshop, IndabaX-AI4D innovation grants 2020, ICLR 2020.

Volunteer: ICLR 2020, NeurIPS 2019 WiML workshop.

Admissions: Cornell CS PhD applicants starting in Fall 2020.

Organizing: **DeepBayes** summer school 2018, 2019.

## Teaching

Sep 2020–Dec 2020 Teaching Assistant, **Cornell University**.
ORIE 4350, Introduction to Game Theory

Sep 2017–Dec 2017 Practice Lecturer, **Higher School of Economics**.
Machine Learning classes for 3rd year CS students

Jul 2017 Teaching Assistant, **Coursera**.
Quizzes and assignments for Bayesian Methods for Machine Learning course.
The specialization of the course received Coursera Outstanding Educator Award

## Additional education

Aug 2020 (Virtual) DL and RL Summer School, Mila, Canada.

Jul 2019 Machine Learning Summer School, Gatsby Unit, UK.

Jul 2019 Reinforcement Learning Summer School, INRIA, France.

Mar 2017 Spring school "Structural Inference", Humboldt University, Germany.

Feb 2017 Winter school on Reinforcement Learning, MIPT, Russia.